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5: //Date: 18/1/2024
6:
7: #include<iostream>
8: #include<fstream>
9: #include<string>
10: #include<iomanip>
11: using namespace std;
12:
13: void readFile(fstream& infile, int salesJan[], int salesFeb[], int salesMar[], int salesApr[], int salesMay[], int salesJun[], int salesJul[], int salesAug[], int salesSept[], int salesOct[], int salesNov[], int salesDec[], string store[]);
14: float grandTotalSales(int salesJan[], int salesFeb[], int salesMar[], int salesApr[], int salesMay[], int salesJun[], int salesJul[], int salesAug[], int salesSept[], int salesOct[], int salesNov[], int salesDec[]);
15: float totalpermonth(int sales[]);
16: void highestSale(int salesJan[], int salesFeb[], int salesMar[], int salesApr[], int salesMay[], int salesJun[], int salesJul[], int salesAug[], int salesSept[], int salesOct[], int salesNov[], int salesDec[], string store[], float& highestsale, string& highestbranch, string& highestmonth);
17: void lowestSale(int salesJan[], int salesFeb[], int salesMar[], int salesApr[], int salesMay[], int salesJun[], int salesJul[], int salesAug[], int salesSept[], int salesOct[], int salesNov[], int salesDec[], string store[], float& lowestsale, string& lowestbranch, string& lowestmonth);
18:
19: int main()
20: {
21:     fstream infile("sales2014.txt", ios::in);
22:     fstream outfile("salesreport.txt", ios::out);
23:     const int a = 12;
24:     int salesJan[a],salesFeb[a], salesMar[a],salesApr[a],salesMay[a];
25:     int salesJun[a], salesJul[a],salesAug[a],salesSept[a],salesOct[a],salesNov[a],salesDec[a];
26:     string store[5];
27:     float grandtotal, totalJan, totalFeb, totalMar, totalApr, totalMay, totalJun, totalJul, totalAug, totalSept, totalOct, totalNov, totalDec;
28:     float totalavg, highestsale = 0, lowestsale = 94000, totalsalesbranch[5];
29:     string highestbranch, highestmonth, lowestbranch, lowestmonth;
30:
31:     if(!infile)
32:     {
33:         cout<<"error when opening files";
34:         exit(1);
35:     }
36:     readFile(infile,salesJan,salesFeb,salesMar,salesApr,salesMay,salesJun,salesJul,salesAug,salesSept,salesOct,salesNov,salesDec,store);
37:     grandtotal = grandTotalSales(salesJan,salesFeb,salesMar,salesApr,salesMay,salesJun,salesJul,salesAug,salesSept,salesOct,salesNov,salesDec);
38:     totalJan = totalpermonth(salesJan);
39:     totalFeb = totalpermonth(salesFeb);
40:     totalMar = totalpermonth(salesMar);
41:     totalApr = totalpermonth(salesApr);
42:     totalMay = totalpermonth(salesMay);
43:     totalJun = totalpermonth(salesJun);
44:     totalJul = totalpermonth(salesJul);
45:     totalAug = totalpermonth(salesAug);
46:     totalSept = totalpermonth(salesSept);
47:     totalOct = totalpermonth(salesOct);
48:     totalNov = totalpermonth(salesNov);
49:     totalDec = totalpermonth(salesDec);
50:
51:     totalavg = grandtotal / 12;
52:
53:     highestSale( salesJan, salesFeb, salesMar, salesApr,salesMay, salesJun, salesJul, salesAug,salesSept,salesOct,salesNov,salesDec,highestsale,store,highestbranch,highestmonth);
54:     lowestSale(salesJan, salesFeb, salesMar,salesApr,salesMay, salesJun, salesJul,salesAug,salesSept,salesOct,salesNov,salesDec,lowestsale,store,lowestbranch, lowestmonth);

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55:
56:     for(int i=0;i<5;i++)
57:     {
58:         totalsalesbranch[i]= (salesJan[i]+salesFeb[i]+salesMar[i]+salesApr[i]+salesMay[i]+salesJun[i]+salesJul[i]+salesAug[i]+salesSept[i]+salesOct[i]+salesNov[i]+salesDec[i])*10
59:     }
60:
61:     outfile<<fixed<<setprecision(2);
62:     outfile<<"Total of sales over all stores: RM "<<right<<setw(10)<<grandtotal<<endl;
63:     outfile<<"Average sales per month: RM"<<right<<setw(10)<<totalavg<<endl;
64:     outfile<<"\n\nThe highest sales:\n-----\nStore:"<<highestbranch<<"\nMonth: "<<highestmonth<<"\nSales: RM"<<right<<setw(10)<<highestsale;
65:     outfile<<"\n\nThe lowest sales:\n-----\nStore:"<<lowestbranch<<"\nMonth: "<<lowestmonth<<"\nSales: RM"<<right<<setw(10)<<lowestsale;
66:     outfile<<"\n\n\nTotal sales by month:\nMonth\tSales\n-----\t-----";
67:     outfile<<"\nJan\tRM"<<right<<setw(10)<<totalJan;
68:     outfile<<"\nFeb\tRM"<<right<<setw(10)<<totalFeb;
69:     outfile<<"\nMar\tRM"<<right<<setw(10)<<totalMar;
70:     outfile<<"\nApr\tRM"<<right<<setw(10)<<totalApr;
71:     outfile<<"\nMay\tRM"<<right<<setw(10)<<totalMay;
72:     outfile<<"\nJun\tRM"<<right<<setw(10)<<totalJun;
73:     outfile<<"\nJul\tRM"<<right<<setw(10)<<totalJul;
74:     outfile<<"\nAug\tRM"<<right<<setw(10)<<totalAug;
75:     outfile<<"\nSep\tRM"<<right<<setw(10)<<totalSept;
76:     outfile<<"\nOct\tRM"<<right<<setw(10)<<totalOct;
77:     outfile<<"\nNov\tRM"<<right<<setw(10)<<totalNov;
78:     outfile<<"\nDec\tRM"<<right<<setw(10)<<totalDec;
79:     outfile << "\n\nTotal Sales by store: " << endl;
80:     outfile << "Store \t\tTotal Sales" << endl;
81:     outfile << "----- \t\t-----" << endl;
82:     for(int i=0;i<5;i++)
83:     {
84:         outfile<<left<<setw(15)<<store[i]<<" RM "<<right<<setw(10)<<totalsalesbranch[i]<<endl;
85:     }
86:     outfile<<"\n\nProfitable stores:\n-----";
87:     for(int i=0;i<5;i++)
88:     {
89:         if(totalsalesbranch[i]>=600000)
90:             outfile<<endl<<store[i];
91:     }
92:
93:     infile.close();
94:     return 0;
95: }
96:
97: void readFile(fstream& infile, int salesJan[], int salesFeb[], int salesMar[], int salesApr[], int salesMay[], int salesJun[], int salesJul[], int salesAug[], int salesSept[], int salesOct[], int salesNov[], int salesDec[])
98: {
99:     for(int i=0; i<5; i++)
100:    {
101:        infile>>salesJan[i]>>salesFeb[i]>>salesMar[i]>>salesApr[i]>>salesMay[i];
102:        infile>>salesJun[i]>>salesJul[i]>>salesAug[i]>>salesSept[i]>>salesOct[i]>>salesNov[i]>>salesDec[i];
103:        getline(infile,store[i]);
104:    }
105: }
106:
107: float grandTotalSales(int salesJan[], int salesFeb[], int salesMar[], int salesApr[], int salesMay[], int salesJun[], int salesJul[], int salesAug[], int salesSept[], int salesOct[], int salesNov[], int salesDec[])
108: {

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109:     float grandsales = 0;
110:
111:     for(int i=0; i<5; i++)
112:     {
113:         grandsales = grandsales + salesJan[i]+salesFeb[i]+salesMar[i]+salesApr[i]+salesMay[i]+salesJun[i]+salesJul[i]+salesAug[i]+salesSept[i]+salesOct[i]+salesNov[i]+salesDec[i]
114:     }
115:     grandsales = grandsales*1000;
116:     return grandsales;
117: }
118:
119: float totalpermonth(int sales[])
120: {
121:     float totalsales = 0;
122:     for(int i=0; i<5; i++)
123:     {
124:         totalsales = totalsales + sales[i];
125:     }
126:     totalsales = totalsales*1000;
127:     return totalsales;
128: }
129:
130: void highestSale(int salesJan[], int salesFeb[], int salesMar[], int salesApr[], int salesMay[], int salesJun[], int salesJul[], int salesAug[], int salesSept[], int salesOct[],
131: {
132:     float jan,feb,mar,apr,may,jun,jul,aug,sep,oct,nov,dec;
133:
134:     for(int i=0; i<5; i++)
135:     {
136:         jan = salesJan[i]*1000;
137:         feb = salesFeb[i]*1000;
138:         mar = salesMar[i]*1000;
139:         apr = salesApr[i]*1000;
140:         may = salesMay[i]*1000;
141:         jun = salesJun[i]*1000;
142:         jul = salesJul[i]*1000;
143:         aug = salesAug[i]*1000;
144:         sep = salesSept[i]*1000;
145:         oct = salesOct[i]*1000;
146:         nov = salesNov[i]*1000;
147:         dec = salesDec[i]*1000;
148:
149:         if(jan>highestsale)
150:         {
151:             highestsale = jan;
152:             highestbranch = store[i];
153:             highestmonth = "Jan";
154:
155:         }
156:         if(feb>highestsale)
157:         {
158:             highestsale = feb;
159:             highestbranch = store[i];
160:             highestmonth = "Feb";
161:         }
162:         if(mar>highestsale)

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163: {
164:     highestsale = mar;
165:     highestbranch = store[i];
166:     highestmonth = "Mar";
167: }
168: if(apr>highestsale)
169: {
170:     highestsale = apr;
171:     highestbranch = store[i];
172:     highestmonth = "Apr";
173: }
174: if(may>highestsale)
175: {
176:     highestsale = may;
177:     highestbranch = store[i];
178:     highestmonth = "May";
179: }
180: if(jun>highestsale)
181: {
182:     highestsale = jun;
183:     highestbranch = store[i];
184:     highestmonth = "Jun";
185: }
186: if(jul>highestsale)
187: {
188:     highestsale = jul;
189:     highestbranch = store[i];
190:     highestmonth = "Jul";
191: }
192: if(aug>highestsale)
193: {
194:     highestsale = aug;
195:     highestbranch = store[i];
196:     highestmonth = "Aug";
197: }
198: if(sep>highestsale)
199: {
200:     highestsale = sep;
201:     highestbranch = store[i];
202:     highestmonth = "Sep";
203: }
204: if(oct>highestsale)
205: {
206:     highestsale = oct;
207:     highestbranch = store[i];
208:     highestmonth = "Oct";
209: }
210: if(nov>highestsale)
211: {
212:     highestsale = nov;
213:     highestbranch = store[i];
214:     highestmonth = "Nov";
215: }
216: if(dec>highestsale)
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217:     {
218:         highestsale = dec;
219:         highestbranch = store[i];
220:         highestmonth = "Dec";
221:     }
222: }
223:
224: }
225:
226: void lowestSale(int salesJan[], int salesFeb[], int salesMar[], int salesApr[], int salesMay[], int salesJun[], int salesJul[], int salesAug[], int salesSept[], int salesOct[], int salesNov[], int salesDec[])
227: {
228:     float jan,feb,mar,apr,may,jun,jul,aug,sep,oct,nov,dec;
229:
230:     for(int i=0; i<5; i++)
231:     {
232:         jan = salesJan[i]*1000;
233:         feb = salesFeb[i]*1000;
234:         mar = salesMar[i]*1000;
235:         apr = salesApr[i]*1000;
236:         may =
237:         jun = salesJun[i]*1000;
238:         jul = salesJul[i]*1000;
239:         aug = salesAug[i]*1000;
240:         sep = salesSept[i]*1000;
241:         oct = salesOct[i]*1000;
242:         nov = salesNov[i]*1000;
243:         dec = salesDec[i]*1000;
244:
245:         if(jan<lowestsale)
246:         {
247:             lowestsale = jan;
248:             lowestbranch = store[i];
249:             lowestmonth = "Jan";
250:
251:         }
252:         if(feb<lowestsale)
253:         {
254:             lowestsale = feb;
255:             lowestbranch = store[i];
256:             lowestmonth = "Feb";
257:         }
258:         if(mar<lowestsale)
259:         {
260:             lowestsale = mar;
261:             lowestbranch = store[i];
262:             lowestmonth = "Mar";
263:         }
264:         if(apr<lowestsale)
265:         {
266:             lowestsale = apr;
267:             lowestbranch = store[i];
268:             lowestmonth = "Apr";
269:         }
270:         if(may<lowestsale)

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271: {
272:     lowestsale = may;
273:     lowestbranch = store[i];
274:     lowestmonth = "May";
275: }
276: if(jun<lowestsale)
277: {
278:     lowestsale = jun;
279:     lowestbranch = store[i];
280:     lowestmonth = "Jun";
281: }
282: if(jul<lowestsale)
283: {
284:     lowestsale = jul;
285:     lowestbranch = store[i];
286:     lowestmonth = "Jul";
287: }
288: if(aug<lowestsale)
289: {
290:     lowestsale = aug;
291:     lowestbranch = store[i];
292:     lowestmonth = "Aug";
293: }
294: if(sep<lowestsale)
295: {
296:     lowestsale = sep;
297:     lowestbranch = store[i];
298:     lowestmonth = "Sep";
299: }
300: if(oct<lowestsale)
301: {
302:     lowestsale = oct;
303:     lowestbranch = store[i];
304:     lowestmonth = "Oct";
305: }
306: if(nov<lowestsale)
307: {
308:     lowestsale = nov;
309:     lowestbranch = store[i];
310:     lowestmonth = "Nov";
311: }
312: if(dec<lowestsale)
313: {
314:     lowestsale = dec;
315:     lowestbranch = store[i];
316:     lowestmonth = "Dec";
317: }
318: }
319: }
```